

Solution to Problem #0 on HW#1

(Part (a)): The answer is $|\det A|$.

Following Part (ii) of the hint, let us start with the special case $|\det A| = 1$: First observe that the classical adjoint formula for the determinant tells us that the entries of A^{-1} are polynomials in the entries of A , divided by $\det A$. So $|\det A| = 1$ clearly implies that $A^{-1} \in \mathbb{Z}^{n \times n}$. So then, the map $m : (\mathbb{C}^*)^n \rightarrow (\mathbb{C}^*)^n$ defined by $m(x) := x^A$ is an automorphism¹: It has the natural inverse $m^{-1}(x) := x^{A^{-1}}$, since $m(m^{-1}(x)) = (x^{A^{-1}})^A = x^{A^{-1}A} = x^{I_n} = (x_1, \dots, x_n)$, and $m^{-1}(m(x)) = x$ similarly, thanks to the algebraic identity observed in the statement of the problem. (We use I_n for the $n \times n$ identity matrix.) In particular, when $|\det A| = 1$, we see that $x^A = c$ must have a unique solution in $\mathbb{C}^n$ which in fact lies in $(\mathbb{C}^*)^n$ since $c \in (\mathbb{C}^*)^n$.

Note: matrices in $\mathbb{Z}^{n \times n}$ with determinant ± 1 are usually called *unimodular*.

To prove the general case, let $UA = H$ (with $U, H \in \mathbb{Z}^{n \times n}$, U unimodular, and H upper-triangular) be the Hermite Factorization of A . From our last paragraph, we know that the map $m(x) = x^U$ is an automorphism of $(\mathbb{C}^*)^n$. So, if we make the change of variables $x := y^U$, the new system $y^H = c$ will have the same number of roots in $(\mathbb{C}^*)^n$ as the original system $x^A = c$: Simply observe that $x^A = (y^U)^A = y^{UA} = y^H$. Note also that $|\det A| = 1 \cdot |\det U| = |\det U| |\det A| = |\det UA| = |\det H|$.

To count the roots of $y^H = c$, simply note that the equations written in full are:

$$\begin{aligned} y_1^{h_{1,1}} &= c_1 \\ y_1^{h_{1,2}} y_2^{h_{2,2}} &= c_2 \\ &\vdots \\ y_1^{h_{1,n}} \cdots y_n^{h_{n,n}} &= c_n \end{aligned}$$

since H is upper-triangular. So solving this system can be done simply by solving the first equation for y_1 , plugging each such y_1 into the second equation, solving for the values of y_2 corresponding to each value of y_1 , etc. Note that each $h_{i,i}$ is nonzero, since $|\det H| = |\det A| \neq 0$. Clearly then, the number of solutions $(x_1, \dots, x_n) \in (\mathbb{C}^*)^n$ is $|h_{1,1}| |h_{2,2}| \cdots |h_{n,n}| = |h_{1,1} \cdots h_{n,n}| = |\det H|$. Since $|\det A| = |\det H|$, we are done.

Note: The answer to this problem in fact remains the same if we instead assume $c \in (K^*)^n$ and ask for solutions in K^n , where K is *any* algebraically closed field. ■

(Part (b)): The answer is $2^{n-r_2(A)}$, where $r_2(A)$ is the rank, over the finite field $\mathbb{F}_2$, of the mod 2 reduction of the matrix A .

To prove our answer, let $UAV = S$ be a Smith Factorization of A , and let $x := y^U$. Similar to before, the map $m(x) := x^U$ defines an automorphism of $(\mathbb{R}^*)^n$. So the number of roots in $(\mathbb{R}^*)^n$ of $x^A = c$ is the same as the number of roots of $y^{UA} = c$, and exponentiating both sides of the last equation by V leaves the roots unchanged. In other words, $y^{UAV} = c^V$ has the same number of roots in $(\mathbb{R}^*)^n$ as $x^A = c$. Since $y^{UAV} = y^S$, we are then left with the diagonal system

$$\begin{aligned} y_1^{s_{1,1}} &= c_1 \\ y_2^{s_{2,2}} &= c_2 \\ &\vdots \\ y_n^{s_{n,n}} &= c_n \end{aligned}$$

1. i.e., an automorphism of the group $(\mathbb{C}^*)^n$ with respect to coordinate-wise multiplication (and thus a bijection in particular).

where $s_{i,i}$ is the i^{th} diagonal entry of S (from upper-left to bottom-right). Now observe that, for any $d \in \mathbb{Z} \setminus \{0\}$ and $\gamma > 0$, the equation $z^d = \gamma$ has exactly 1 or 2 roots according as d is odd or even. So the number of solutions of $y^S = c^V$ is simply 2 to the number of even $s_{i,i}$. To conclude, simply note that Smith Factorization $UAV = S$ (which is an identity over $\mathbb{Z}$) continues to hold mod 2. In particular, $r_2(A) = r_2(S)$, and $r_2(s)$ is simply the number of odd diagonal entries of S . So we are done. ■

(Part (c)): The answer is $\boxed{\gcd(d, q-1)}$. The proof hinges on some facts that should be familiar to us by now (from any reasonable first year abstract algebra class):

(1) The multiplicative group $\mathbb{F}_q^*$ is cyclic.

(2) If g is a generator of $\mathbb{F}_q^*$ then the order of g^d is $(q-1)/\gcd(d, q-1)$.

Now, $x^d = c$ having a solution means that c is a d^{th} power. So the solutions of $x^d = c$ form a coset of the subgroup of $\mathbb{F}_q^*$ consisting of d^{th} powers. So the number of solutions of $x^d = c$ in $\mathbb{F}_q^*$ is the same as the cardinality of the subgroup of $\mathbb{F}_q^*$ defined by $x^d = 1$.

Now, from Facts (1) and (2), the map $m(x) := x^d$ sends $\mathbb{F}_q^*$ surjectively to a subgroup of order $(q-1)/\gcd(d, q-1)$. So the kernel of m has cardinality $\gcd(d, q-1)$ and we are done. ■